

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

SA1 - Health and Care

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

- Q. 1)** You are an actuary working in the actuarial consultancy firm. ABC Hospital Group has approached your company and requested to design and price a new product suite to give patients certainty and transparency when planning for a surgery or medical procedure.

ABC Hospital Group intends to target individuals who do not have a health insurance policy. Individuals will pay upfront for this product and in case of any hospitalization event, full cost of treatment will be covered under this product. Customers are not required to pay any out-of-pocket expenditure.

The price of this new product offered by the ABC Hospital Group must include the full cost related to hospitalization including room and nursing charges, doctor fees, specialist fees, any prescribed tests, medication and post-operative follow-up, if required. In the event of any related complications, those costs are also covered under this product.

- a) Discuss product design features and how it might support demand for such a product. (15)
- b) Briefly outline the features of the surgeries or medical procedures that would make them suitable for cover under such a prepaid product. (10)
- c) Explain how you would determine a price for such a product that would satisfy the benefit design (15)

[40 Marks]

- Q. 2)** You are working in Indian branch of a reinsurance company which is part of a large international reinsurance group. You are calculating the IBNR reserve as at 31 December 2025 for the local branch.

- a) List 4 factors that you should consider when grouping the data to calculate the claims reserve for the entire inwards reinsurance book of your company. (2)
- b) Outline additional data issues that you should consider when reserving for inwards proportional reinsurance business compared to a non-reinsurer reserving for insurance business. (2)

Your team member has provided last three years' information for the Critical Illness proportional treaty which is running from last 5 years.

Underwriting Year	Written Premium ₹ In Lakhs	Cumulative Claims Incurred as at 31/12/2025 ₹ In Lakhs	Percentage of claims incurred developed as at 31/12/2025
2022-2023	1,200	945	85%
2023-2024	1,875	1,080	65%
2024-2025	3,300	750	30%

You have also been provided with the following additional information:

1. The reinsurance treaties were written on a risks-attaching basis.
2. The above figures are gross of any applicable retrocessions.
3. Treaty covers all the underlying Critical Illness insurance policies were single premium policies which commenced on 1 April and provide cover for 3 years on average.
4. The percentages of claims incurred developed were calculated from the underwriting year claims development triangles using the basic chain ladder method.
5. Your company's reserving policy calculates the IBNR for reinsurance business by underwriting year.

- c) Explain the difference between the ultimate claims estimated using the chain ladder method applied to accident year claims triangles versus underwriting year claims triangles. (2)
- d) Explain why earned premium and not written premium is used to calculate the IBNR reserve in the formula above for the 2024-2025 underwriting year. (2)
- e) Calculate the aggregated IBNR reserve for the three underwriting years 2022-2023 to 2024-2025 based on your company's reserving policy explained above. (5)
- f) Outline issues you should consider before finalising the IBNR reserve for the local engineering proportional treaty reinsurance book. (2)

[15 Marks]

Q.3) A life insurance company in your country has written significant volumes of only conventional income protection business over the past few years.

- a) Explain how income protection business meets the needs of customers. (15)
- b) Explain why the company will want to monitor the experience and profitability of this business on an ongoing basis. (8)
- c) Describe the methods that the company is likely to use to monitor the existing business. (10)
- d) Give, with reasons, suitable asset classes for the company's investments. You should also indicate, with reasons, the further information you would ideally like to have to refine your asset mix. (12)

[45 Marks]
